

Pre-screening

Participant	018
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	Yes	No
Do you or have you had a history of migraine headaches?	☐	☒
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐	☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Female
3. How old are you?	42
4. What is the highest level of education you have completed?	☒ Some college/Associates/Technical school training
5. Which option best describes your current home?	☒ It is rented for money by you or someone in household.
6. How do you pay for healthcare?	☒ Government funding (Medicaid, Medicare, Ryan White, VA)
8. Which best corresponds to your current work situation?	☒ Currently in school.
9. What is your combined family income for the past 12 months (from all sources)	☒ \$5,000-\$19,999

Timeline Followback

Visit 1

Visit 1 Date

06-17-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 2

Visit 2 Date

07-01-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 3

Visit 3 Date

07-16-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 4

Visit 4 Date

07-29-2025

Visit 5

Visit 5 Date

08-05-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Phone

Participant # _____

18

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☐	☒	☐	☐	☐
2. The time I spent in this app just slipped away.	☐	☐	☒	☐	☐
3. I was absorbed in the app task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☒	☐	☐
8. The app is aesthetically appealing.	☐	☐	☒	☐	☐
9. The app appealed to my senses.	☐	☐	☐	☐	☒
10. Using this app was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were videos easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

AR

Participant ID _____

018

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space | ☒ Somewhat Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Disagree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☐	☒	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☐	☒	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☒	☐	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☐	☐	☐	☐	☒
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☒	☐	☐	☐	☐

8. The augmented reality is aesthetically appealing.	☐	☐	☒	☐	☐
9. The augmented environment appealed to my senses.	☐	☒	☐	☐	☐
10. Using this system was worthwhile.	☐	☐	☒	☐	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☒	☐	☐	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

VR

Participant ID _____

18

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Government Funding (Medicaid, Medicare, Ryan White, VA) |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Disagree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Somewhat unaware |
| 8. I was not aware of my real environment. | ☒ Somewhat Agree |
| 9. I still paid attention to my real environment. | ☒ Completely Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11. How real did the virtual world seem to you? | ☒ Very real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Agree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☐	☐	☒	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☐	☐	☒	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☒	☐	☐
2. The time I spent in the virtual world just slipped away.	☐	☐	☐	☐	☒
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☐	☒

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☐	☒
9. The virtual environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

MR

Participant ID _____

018

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire MR

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely disagree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat disagree |
| 3. I felt like I was perceiving pictures. | ☒ Neither Agree nor Disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat disagree |
| 6. I felt present in the virtual space. | ☒ Completely disagree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely disagree |
| 11. How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Neither Agree nor Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☒	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☒	☐	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☐	☒	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☒	☐	☐	☐	☐
2. The time I spent in mixed reality just slipped away.	☐	☒	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☒	☐	☐	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The blended environment appealed to my senses.	☒	☐	☐	☐	☐
10. Using this system was worthwhile.	☐	☐	☒	☐	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is rented for money by you or someone in household
2. How do you pay for healthcare?	☒ Government Funding (Medicaid, Medicare, Ryan White, VA)
3. Which best corresponds to your current work situation?	☒ Not working or looking for work.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	enjoyed VR the most due to it being the most immersive. can definitely tell it's virtual and not real, but feels like you could potentially put yourself there if you were comfortable enough. did not enjoy MR much and felt that it was difficult to focus on the tasks. disliked AR the most and felt like it was incredibly difficult to focus on the task while being able to see the background.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	yes - only VR. AR could be very aggravating for people and feels like VR could be beneficial for individuals that could be having a very difficult day with cravings at a 10. helpful to take a moment through VR and get their minds off of what is going on.
3. How would you describe the iPAL application, in terms of a cravings management tool?	believes that it is a good cravings management tool. the questions make you aware of what's going on in your life on a daily basis and how you're feeling and what's causing those feelings. could be helpful for feelings management as well and could potentially help with linking a cause behind the feelings.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - enjoyed the videos a lot. whenever someone is made aware of those feelings and cravings they are experiencing and understanding how to manage them, this could be helpful in recovery. participant feels that sometimes they have difficulty with remembering to check in with themselves and what's going on. HRV - felt like the exercise was too long. feels like it may have been a good length to keep someone grounded if coordinator wasn't in the room during the intervention.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - there are multiple pathways for a reason and participant believes that if the application were presented as an option, it could be very beneficial to others in recovery. should be utilized when people are coming into recovery and getting out of treatment.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - would envision utilizing twice per week